Annexure-I

Title of the work

Market basket analysis

Jammu & Kashmir Bank

A training report

Submitted in partial fulfillment of the requirements for the award of degree of

MBA(Business Analytics)

Submitted to

LOVELY PROFESSIONAL UNIVERSITY

PHAGWARA, PUNJAB

From 25/06/24 to 09/08/24

SUBMITTED BY

Name of student: Muwahid Munshi

Registration Number: 12306969.

Annexure-II: Student Declaration

To whom so ever it may concern

I, **Muwahid Munshi**, bearing registration number **12306969**, hereby declare that the work done by me on "**Market Basket Analytics**" from **June**, **2024** to **August**, **2024**, is a record of original work for the partial fulfillment of the requirements for the award of the degree, **MBA(business Analytics).**

Muwahid Munshi (12306969)

Dated: 31-08-24

TRAINING CERTIFICATE FROM THE ORGANISATION

The Jammu & Kashmir Bank Limited
Corporate Headquarters
M.A Road, Srinagar 190001
Kashmir, India
T: +91 (0)194 2713139
F: +91 (0)194 250 3678
E: Ps.Training@jkbmail.com
W: www.jkbank.net

J&K Bank
Serving To Empower

Human Resources

Ref: JKB/HR/TRNG/2024-579

August 31, 2024

TO WHOM IT MAY CONCERN

This is to certify that Mr. Muwahid Munshi, S/O Mr. Zaffar Altaf Munshi, pursuing MBA (Business Analytics) from "LPU, Mittal School of Business" has completed his Project Work/Internship at J&K Bank on the topic "Market Basket Analysis", from 25th June, 2024 to 09th August, 2024.

His work and conduct has remained excellent.

We wish him all success.

J&K Bank
HRD Division
Corporate Headquarters
M. A. Road, Srinagar

HR-Trainings

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INTRODUCTION TO BANKING:

Banking is nothing but a service. Banks are business organizations selling banking services. Banks continuously reassess how a customer views bank services, what are new and emerging customer aspirations and how these can be satisfied.

ORIGIN:

Since the banking activities were started in different periods in different countries, there is no unanimous view regarding the origin of the word bank. The word bank is derived from the French word 'Banc' or 'Bancus', which means a bench. In fact the early Jews in Lombardy transacted their banking business sitting on benches. When the business ailed, the benches were broken and hence the word bankrupt came in to vogue.

DEFINITION:

The Indian Banking Companies Act, 1949 section 5(b), defines banking as "accepting for purpose of lending or investing of deposits from the public, repayable on demand or otherwise and withdrawals by cheque, drafts, and orders or otherwise".

Banks are backbone of our society. A bank must meet the financial needs of a customer, by acting as a custodian of his assets, providing credit facilities and assisting him to speedily put through financial transaction of one type or another. Banking when you come to think of its people. It is not figures, files and ledgers. Bank services needs considerable improvement on an emergent basis. The time has come for banks to look inward to find out what is the nature and quality of the products they sell, what is the product demanded by the customer.

It would be unrealistic today to believe that banks are mere financial institutions, working for profit. Banks essentially are now social organizations, regarding financial services to sub serving the socio-economic objectives of the society.

IN INDIA:

India has a system of indigenous banking from very early times, though it was not similar to banking of modern times. There is evidence to show that money lending existed even during the Vedic period. With the advent of the English traders in the seventeenth century and the establishment of trading centers by the East India Company encouraged the establishment of what were known as the agency houses? The trading firms which undertook banking operations for the benefit of their constituents. Some of the houses established during the period were Alexander and Co. and Ferguson & Co. There were also Presidency Banks, Joint stock banks and Exchange banks which took up gradually one after the other.

IMPERIAL BANK:

The Presidency Banks referred to above were amalgamated into the Imperial Bank of India, which was brought into existence on 27th January 1921 by the Imperial Bank of India Act 1920. This Act, however, gave the bank no power to issue notes and this left out without control over the currency of the country. But it was allowed to hold Government balances and to manage public debt and clearing houses till the establishment of Reserve Bank Of India in 1935 which apart from taking over all these functions from the Imperial Bank of India, was given the privilege of acting as an agent of the latter in places where it had no branches.

COMMERCIAL BANKS:

Amongst the banking institutions in the organized sector, the commercial banks initially were established as corporate bodies with shareholdings by private individuals, but subsequently there has been a drift towards central ownership and control. Today 27 banks constitute the strong public sectors in Indian Commercial Banking. Up to late 60's Commercial Banks are mainly engaged in financing organized trade, commerce &industry, since then they are actively participating in financing agriculture, small-scale business and small borrowers also.

FUNCTIONS OF COMMERCIAL BANKS:

Commercial banks perform several crucial functions, which may be classified into two categories:

(a) Primary functions

(b) Secondary functions

- **PRIMARY FUNCTIONS:**

Primary banking functions of the commercial banks include:

- Acceptance of deposits from public.
- Lending funds.
- Use of cheque system and
- Remittance of funds.

- **SECONDARY FUNCTIONS:**

Commercial banks perform a multitude of other non-banking functions, which may be classified as:

- Agency service.
- General utility services.

NATIONALIZATION OF COMMERCIAL BANKS:

The major historical event in the history of banking in India after independence is undoubtedly the nationalization of the 14 major banks on 19th July 1969. In 1980, six more private sector banks are nationalized extending the public domain further over the banking sector. Nationalization was deemed as a major step in achieving the socialistic pattern of society. The nationalized banks were to increase lending to areas of importance to the government and to use their resources for sub-serving the common good. A detailed scheme of objectives, regulations, management, etc was drawn up for these banks.

Nationalization was recognition of the potential of the banking system to promote broader economic objectives. The banks had to reach out and expand their networks so that the concept of mass banking was given importance over class banking

THE RESERVE BANK OF INDIA [RBI]:

The Central Bank of India called the Reserve Bank of India was constituted under the Reserve Bank of India Act, 1934 to regulate the issue of bank notes and keeping of reserves with a view to securing monetary stability in India and generally to operate the currency and credit system of India to its advantage. Amongst its multifarious functions affecting the Indian Financial System, the RBI regulates and prohibits the issue of prospectus or advertisement soliciting deposits of money, regulates the functioning of non-banking institutions and transacts Government business. Its regulatory involvement in the Indian Capital Markets is primarily of debt management through primary dealers, foreign exchange control and liquidity support to market participants. The RBI regulates participants in the securities markets when a foreign transaction is involved. Transactions that include Indian issuers issuing of security outside India, such as GDRs and ADRs, and Financial Institutional Investors (FIIs) or Foreign Brokers selling, buying or dealing in Indian Securities need the permission of RBI.

As the central banking authority of India, the Reserve Bank of India performs the following traditional functions of the central bank:

- It provides currency and operates as the clearing system for the banks.
- It formulates and implements monetary and credit policies.
- It functions as the banker's bank.
- It supervises the operations of credit institutions.
- It regulates foreign exchange transactions.
- It moderates the fluctuations in the exchange value of the rupee.

In addition to the traditional function of the central banking authority, the Reserve Bank of India performs several functions aimed at developing the Indian financial system:

- It seeks to integrate the unorganized financial sector with the organized financial sector.
- It encourages the extension of the commercial banking system in the rural areas.

- It influences the allocation of credit.
- It promotes the development of new institutions.

REVIEW OF INDIAN BANKING SCENE:

Banking sector reforms have brought about considerable shift in the progress of different banks. Credit has once again gained importance. Technology has brought banking to the very doorstep of customers. With the coming of foreign banks, competition has increased and they are models for the other banks in terms of service level, banking convenience technology and staff productivity. In order to keep them alive in the race for survival and in order to increase profitability, banks have been forced to become active in offering services to customer. In fact globally the dominance of internet and cyber banking has changed the banking scenario.

ROLE OF BANKS IN ECONOMIC DEVELOPMENT:

Commercial banks are playing a crucial role in the economic development of the country. In fact without the development of commercial banks in 18th century, industrial revolution would not have taken place in England at all. It is also true that economic development of country depends entirely on the development of sound commercial banking.

Banks provide short term loans which serve as capital for industrial establishment. Without capital it is impossible to start an industry. After starting the industry, the banks provide the industrialists necessary working capital. Thus by providing with investment capital and short term working capital, the banks encourage industrial advancement in the country.

Banks extend credit facilities to industry and trade to develop right type of industry and business. Expansion of credit will provide more funds for the entrepreneurs to start new industries, which results in more employment and income. Commercial banks by providing funds encourage production and cause an increase of national income by means of transferring surplus resources obtained from rural sector. Banks promote capital formation by means of

pooling savings from the people. They mobilize the idle and dormant capital of the community and provide it for investment. Banks can also influence the economy in so many other ways. Banks can regulate the interest rates in the money market by means of regulating the supply of the funds. A cheap money policy with low rate of interest will tend to stimulate economic activity during the period of inflation. A reverse policy is followed during depression.

BANKING SCENARIO:

During the year 2005-2006, the reserve bank of India took several initiatives aimed at improving the prudential regulation. These include stipulating higher provisioning requirement for NPAs included under "doubtful for more than three years" category effective from March 31, 2006, prohibiting banks from investing in unrelated non-SLR securities, advising banks to maintain capital charge for market risk etc. Further several initiatives were also taken during the year aimed at improving the credit delivery to the agricultural and SSI sector. The govt. announced a comprehensive policy envisaging a 30% increase in agriculture credit in 2005-06 and doubling the credit flow to the agriculture sector in three years.

INTRODUCTION TO J&K BANK:-

COMPANY PROFILE

Type : Public.

Traded as : NSE: J&KBANK, BSE: 532209.

Industry : Banking, Financial services.

Founded : October 1, 1938.

Headquarters : Srinagar, Jammu and Kashmir, India.

Key people : Sh Baldev Prakash (Chairman & CEO).

Products : Credit cards, Consumer banking, corporate banking, finance and insurance, mortgage loans, private banking, wealth management ,investment banking

Revenue : 12051 crores INR , Mar 2024

Operating income : ₹6,029.17 crore, Mar 2024

Net income : 1767 crore, Mar 2024

Total assets : ₹1,54,526.59 crore. Mar 2024

OWNER : Government of Jammu and Kashmir (68.18%)

Website : www.jkbank.com

Banks new IDentity:

The new identity for J&K Bank is a visual representation of the banks philosophy and business strategy. The three colored squares represent the three regions of J&K namely Jammu, Kashmir and Ladakh. The counter form created by the interactions of the squares is a falcon with outstretched wings-a symbol of power and empowerment.

Green signifies Growth and Renewal.

Blue conveys Stability and Unity.

Red Represent Energy and Power.

- **VISION**

<u>"To catalyse economic transformation and capitalise on growth."</u>

Our vision is to engender and catalyse economic transformation of Jammu and Kashmir and capitalise from the growth induced financial prosperity thus engineered. The Bank aspires to make Jammu and Kashmir the most prosperous state in the country, by helping to create a new financial architecture for the J&K economy, at the centre of which will be the J&K Bank.

MISSION

Our mission is two-fold: To provide the people of J&K international quality financial service and solutions and to be a super-specialist bank in the rest of the country. The two together will make us the most profitable Bank in the country.

Jammu and Kashmir bank was incorporated on 1st October, 1938 and commenced business in July, 1939. From a small beginning the bank has grown to become a giant with a network of 520 branches spread over the length and breadth of the country.

The J & K Bank is the first state owned bank of the country and 53% of equity is held by the government of Jammu and Kashmir. The bank has a consistent track record of growth and profitability. It has a unique distinction of being banker to the Jammu and Kashmir state government and has also been appointed by RBI as its agency in J & K, responsible for carrying general banking business of the Central government and collection of taxes pertaining to the Central Board of Direct Taxes.

J & K Bank has over the past few years been reporting sustained excellent performance in all the vital areas of business and achieving record-breaking levels in profit generation. However what I feel is more reassuring is that bank is quietly working on a much larger objective of setting a strong foundation that will serve as a launching pad to support the aggressive growth plans in the future. J & K Bank has today acquired significant financial strengths on key bench marks that will give it great confidence in the journey forward.

Today the bank has a status of value driven organization and is always working towards building trust with shareholders, employees, customers, borrowers, regulators and other diverse stakeholders. It has adopted a strategy directed to developing a sound foundation of relationship and trust aimed at achieving excellence which of course comes from the womb of good corporate governance. Good governance is a source of competitive advantage and a critical input for achieving excellence in all pursuits. J & K Bank considers good corporate governance as the sine qua non of a good banking system and has adopted a policy based on all four pillars of good governance –transparency, disclosures, accountability and value, enabling it to practice trusteeship, transparency, fairness and control, leading to stakeholders delight, enhanced shareholder value and ethical corporate citizenship. It also ensures that bank is managed by an independent and highly qualified Board following best globally accepted practices, transparent disclosure and empowerment of shareholders, besides ensuring to meet shareholder aspirations and societal expectations

following the principals of management executive freedom to drive the bank forward without undue restraints but within the framework of effective accountability. The excellence achieved by the bank in its operations stemming from the roots of voluntary good governance has not gone unrecognized and bank has recently bagged three very prestigious award for following fair business practices and commitment to social obligations

HISTORICAL BACK GROUND OF J&K BANK:

Before independence moneylenders use to perform entire banking in the Jammu and Kashmir state that too on exorbitant interest rates. Punjab National Bank, Imperial Bank of India and Grind lay's bank were the banks that use to take deposits but no lending due to their statutory limitations. Money lenders mostly known as Munimji conducted financial transactions in the entire state of Jammu and Kashmir. Under this scenario banks could not ameliorate the financial and social position of the people of the state. To overcome this critical situation, the Maharaja of the state conceived an idea of setting up of a state bank in the state.

After a prolonged exercise and deliberations, the assignment for establishment of "The Jammu and Kashmir Bank limited" was given to the late Sir Sorabji. N. Pochkanwala, the Managing Director of the Central Bank of India. Mr. Pochkanwala formulated a scheme on 24-09-1930, suggesting establishment of a semi state bank with participation in capital by state and the public under the control of state government. Thus the bank was formally incorporated on the 1st of October 1938 and commenced business from 4th of July 1939 at its registered office Residency road Srinagar, Kashmir.

The Jammu and Kashmir bank Limited has been the first of its nature and composition as a state owned bank in the country. The state government besides contributing half of the issued capital also appointed it as its bankers for general banking and treasury business. In its formative years, the bank had to encounter several serious problems particularly around the time of independence when out of its total of ten branches; two branches of Muzaffarabad and Mirpur fell to the other side of the line of control (now Pakistan Administered Kashmir) along with cash and other assets in 1947. However the State government came to its rescue with the assistance of Rs.6.00 Lacks to meet the claims, however the bank fastly over came its difficulties and kept growing. Following the extension of Central Laws to the state of Jammu and Kashmir, the bank was defined as a government company as per the provisions of Indian Companies Act 1956. The

bank had its first full time chairman in 1971, following social central measures in banks. The year 1971 was a turning point for the bank on conferment of scheduled bank status and witnessed remarkable progress in all the vital fields of operations. The bank was declared as "A" class bank by Reserve Bank of India in1976. In recognition of dominant role and exalted performance, Reserve bank of India as its agent for performing the general banking business of the central government especially in maintaining currency chests and collection of taxes.

Performance Highlights for the quarter ended June 30, 2024:

- Net Profit of Rs 415.49 crore for the quarter ended June, 2024 as compared to Net Profit of Rs 326.45 crore during the quarter ended June, 2023.
- Deposits stood at Rs 132573.13 crore as on June, 2024 as compared to Rs 121297.49 crore as on June, 2023.
- Advances stood at Rs 95449.77 crore as on June, 2024 as compared to Rs 84475.63 crore as on June, 2023.
- EPS for the quarter ended June, 2024 at Rs 3.77 compared to Rs 3.16 for the quarter ended June, 2023.
- NIMs for the quarter ended June, 2024 at 3.86 % (annualized) vis-à-vis 3.98% for the quarter ended June, 2023.
- Post tax Return on Assets at 1.08 % (annualized) for the quarter ended June, 2024 compared to 0.94 % for the quarter ended June, 2023.
- Post Tax Return on Average Net-Worth (annualized) for the quarter ended June, 2024 at 14.82 % compared to 14.79 % recorded for the quarter ended June, 2023.
- Cost of Deposits (annualized) for the quarter ended June, 2024 at 4.66 % compared to 4.32 % recorded for the quarter ended June, 2023.

• Yield on Advances (annualized) for the quarter ended June, 2024 stood at 9.50% as compared to 9.30 % for the quarter ended June, 2023.

• Business per Employee and Net Profit per Employee (annualized) were at Rs 18.01 crore and Rs 13.09 lakh respectively for the quarter ended June, 2024 compared to Rs 15.87 crore and Rs 9.98 lakh pertaining to the quarter ended June, 2023.

• Gross and Net NPA's as percentages to Gross and Net Advances as on June, 2024 at 3.91 % and 0.76 % respectively compared to 5.77 % and 1.39 % as on June, 2023.

• NPA Coverage Ratio as on June, 2024 at 91.57 % as compared to 87.55 % as on June, 2023.

• Cost to Income Ratio stood at 61.96 % for the quarter ended June, 2024 as compared to 65.07 % for the quarter ended June, 2023.

• Capital Adequacy Ratio (Basel III) stood at 15.07 % as on June, 2024 which was recorded at 14.83 % as on June, 2023.

Branch/ATM Network

New branches were established there by taking the number of branches to 938 spread over 20 states and one union territory. The area-wise breakup of the branch network (excluding extension counters/ mobile branches and Service branches) on the basis of census 2011 is as under:

AREA	Branches
METRO	172
URBAN	108
SEMI-URBAN	155
RURAL	503

TOTAL	**938**

96 ATMs were commissioned there by taking the number of ATMs to1294

INNOVATIVE PRODUCTS:

To maximize value to its customers, the innovation in products and improving the quality and speed of services is the hallmark of Bank's business strategy. In keeping with this objective, the bank has launched several unique and innovative deposit products, which includes "Mehendi Deposit Scheme", "Recurring plus Deposit Account", "Flexi Deposit Scheme", etc. These schemes have flexibility of depositing variable monthly installments as per the convenience of the depositor. In synchronization with the Bank's new strategy focus on increasing lending and financial deepening of economy in J & K state, a host of new products customized and tailored to the requirements of customers were designed and launched. Some of these products are:

- All purpose Agri-Term Loan.
- JK Bank Term Loan for B.Ed/M.Ed Courses.
- Housing Loan for Leh.
- Tax Saver Term Deposit Scheme.
- Smart Saver.
- NaunihalonKaNayaSavera.
- Used-car Loan
- Value Added Current Accounts.
- JK Bank Dastkar Finance Scheme
- JK Bank Khatamband Scheme
- JK Bank Roshni Finance Scheme
- JK Bank Commercial Premises Finance

- JK Bank Giri Finance Scheme

NAME OF THE BOARD OF DIRECTORS:

• **1. Mr. Baldev Prakash (MD & CEO)** • **Dr. Pawan Kotwal, IAS** • **Mr. Santosh Dattatraya Vaidya** • **Dr. Mandeep K Bhandari** • **Mr. R. K. Chhibber** • **Dr. Rajeev Lochan Bishnoi** • **Mr. Naba Kishore Sahoo** • **Mr. Umesh Chandra Pandey** • **Mr. Anil Kumar Goel** • **Mr. Anand Kumar** • **Mr. Sudhir Gupta** • **Ms. Shahla Ayoub**	**Chairman & MANAGING Director** **Directors**

NEW BUSINESS INITIATIVES:

The various new business initiatives taken by Jammu and Kashmir Bank Limited are as under:

- Innovative financial products.

- Monetizing the Bank's Branch network.
- Third party product distribution.
- Investment banking.
- Venture capital financing.

J&K BANK BRANCHES IN MY STUDY AREA:

Bank	Branch Code	Address
Jammu & Kashmir Bank (JK Bank)	IFSC Code: JAKA0ASHAJI Branch Code: ASHAJI	**Branch:** JK BANK ASHAJIPORA ANANTNAG CITY: SRINAGAR **DISTRICT: ANANTNAG** **State:** JAMMU AND KASHMIR

Machine Learning is rewarding the retail industry in a unique way. It supports the retail sector in all areas, from predicting sales success to locating customers. Market basket analysis (MBA) is one such top retail application of machine learning. It helps retailers know what products people are purchasing together so that the store/website layout can be designed in the same manner. It is mainly done by studying their previous purchasing activity. Companies also leverage it

in cross-selling their products on their online platform. However, it is not only used in the retail sector; fraudulent insurance claims and credit card transactions also make use of it.

Amazon is a great example that leverages this analysis to cross-sell products. These are the products that come under the suggested item list which might interest you along with your current purchase. Your browsing history, what other customers have bought with a given product, and other factors determine which products appear in the suggested category.

Market basket analysis explained

Market basket analysis is a data mining technique that analyzes patterns of co-occurrence and determines the strength of the link between products purchased together. We also refer to it as frequent itemset mining or association analysis. It leverages these patterns recognized in any retail setting to understand the behavior of the customer by identifying the relationships between the items bought by them. To put it simply, market basket analysis helps the retailers know about the products frequently bought together so as to keep those items always available in their inventory.

The source from which these patterns are found is the vast amount of data that is continually collected and stored. With frequent mining of the item set, it becomes easy to discover the correlation between items in huge relational or transactional datasets. It considerably helps in decision-making processes related to cross-marketing, catalog design, and consumer shopping analytics.

You may better comprehend this idea by using the following example.

When people buy green tea, it is evident that they may also buy honey with it. This relationship is depicted as a conditional algorithm, as given below.

IF {green tea} THEN {honey}

It represents that items stated on the right are more likely to be ordered with the items on the left side. Market basket analysis in data mining helps us understand that relationship and how helpful it would be to alter our decisions based on the analysis.

Machine learning engineers develop data-driven strategies based on detailed market basket analysis that further help retailers improve their revenues in the following ways.

- They offer discounts on one of the associated products only.
- The associated products are placed near to each other so that when the customer buys one item, they would be inclined to buy the other item too.

Types of market basket analysis

Market basket analysis comprises the following types.

1. Descriptive market basket analysis

This type of market basket analysis offers actionable insights based on historical data. It is a frequently used approach that does not make any predictions but rates the association using statistical techniques between the products. We also refer to it as unsupervised learning based on the way it is modeled.

2. Predictive market basket analysis

Although "predict" and "analysis" make up the word predictive analysis, it actually works in reverse. It first analyzes and then predicts what the future holds. This type utilizes supervised learning models like regression and classification. It is a valuable tool for marketers even if it is less used than descriptive market basket analysis.

So when we talk about the predictive market basket analysis, it considers items purchased in sequence to evaluate cross-sell. For instance, when a consumer purchases a laptop, they are more likely to buy an extended warranty with it. This analysis thus helps in recognizing those considered items in a sequence so they can be sold together.

It finds application in the retail industry mainly to determine the item baskets that are purchased together.

3. Differential market basket analysis

Differential market basket analysis is a great tool for the competitive analysis that can help you determine why consumers prefer to purchase the same product from a particular platform even when they are labeled with the same price on both platforms.

This decision of the consumers is often based on several factors, as listed below.

- Delivery time
- User experience
- Purchase history between stores, seasons, time periods, and others.

By considering all these factors that are backing the consumers' decision, organizations can benefit from differential market basket analysis. They can make all the parameters fall in accordance with the consumer excel user experience and increase sales on their platform.

Terminologies used in market basket analysis

Here are some terminologies you should keep in mind while working with market basket analysis.

- Itemset: It refers to the set of items that are purchased together by a customer at the same time. By default, we state it as a logical rule with IF and THEN. For instance, IF (Bread, Butter), THEN (Milk). It is also possible that an item set may consist of no items that are usually ignored by all items in the dataset.
- Support count: It is the frequency of a particular item set appearing in the transaction database. It is also stated as a probability. For instance, if milk has a support count of 50 out of a possible 500 transactions, then the probability is 50/500 or 0.1.
- Confidence: It refers to the conditional probability that represents what items have a possibility of being purchased together. It finds application in the product placement strategy intending to increase profitability. For instance, by placing the high-margin items close to the related high-confidence items, retailers can increase the overall sales and revenue on purchases.
- Antecedent: The IF component written on the left-hand side or the item sets within the data are referred to as antecedents.
- Consequent: The THEN component or an item or itemset found in combination with the antecedent is called the consequent.

We can further understand antecedent and consequent with the below example.

$$\{A_i\} \rightarrow \{C_i\}$$

Algorithms used in market basket analysis

Market basket analysis utilizes association rule {IF} - > {THEN} to predict the probability of certain products being purchased together. They count the item frequency occurring together and seek to find associations that occur more than expected.

Some algorithms that leverage these association rules are **AIS, Apriori**, and **SETM**.

Apriori is the commonly cited algorithm by the data scientist that identifies frequent items in the database. It is useful for unsupervised learning and requires no training and thus no predictions. This algorithm is used especially for large data sets where useful relationships among the items are to be determined.

You would be surprised to know that Apriori algorithm leverages a shortcut namely Apriori property. This shortcut states that all items in a frequent itemset must also be frequent. It helps in saving a lot of computational time.

The Apriori algorithm works in two steps that are illustrated below.

- It identifies the itemsets systematically that occur frequently in the dataset and support greater than the pre-specified threshold value.
- Next, it calculates the confidence of all possible rules. However, it only keeps those items states that have confidence greater than a pre-specified threshold.

It is further classified into three components.

- Support
- Lift
- Confidence

Let's understand each one of them with an example of how to calculate market basket.

Assume that a popular eCommerce website makes 100 transactions. Now if we want to calculate the lift, support, and confidence of the two products, say books and pencils, here's how we will proceed.

Let's say there are 10 transactions for books and 8 transactions for pencils and 6 transactions are made for both products.

1. **Support**: It is the total number of transactions made for a particular product divided by the total number of transactions made. Zero represents no support while one represents the highest support. Higher the value of support, the greater the importance of the itemset in the data.

support($A \Rightarrow B$) $=P(A \cup B)$

Support (Books) = Freq (Books)/Total transactions made

Support (Books) = 6/100 = 0.06%

2. **Confidence:** It is the ratio of combined transactions to individual transactions.

confidence($A \Rightarrow B$) $=P(B|A)$

Confidence (Books) = Combined transactions/Individual transaction

Confidence (Books) = 0.06/0.08 = 0.75

3. **Lift:** It is the ratio of the confidence percent to the support percent.

Lift = 0.75/0.10 = 7.5

- If the value of lift < 1, the combination is not bought by consumers frequently.
- If the value of lift >1, the combination is brought frequently by the consumers.
- If the value of lift = 1, then the purchase of antecedent makes no difference on the consequent.

Market basket analysis is used to search for the rules that result in a lift value greater than 1.

Note: Confidence and support can be leveraged to influence the Apriori algorithm. It is done by setting cut-off values to be searched for. For instance, by setting a minimum support value of 0.5 and a confidence value of 0.65, we tell the computer to only report those association rules that are above these cut-off points. It eliminates useless tools that add no value to the decision-making process.

How does market basket analysis work?

Market basket analysis is based on association rule mining which is

IF {}, THEN {} construct

Everything that is within the brackets is referred to as an itemset, which is some form of data. The process initiates from a data set of transactions. Every transaction depicts a group of products that are bought together by the customers and are often referred to as itemsets. These transactions are analyzed through the market basket analysis to identify the rules of the association.

It means that if a customer made a transaction that consisted of bread and butter, then they are likely to purchase milk too. However, before acting on any rule, the store manager or retailer must have sufficient evidence to back up the decision so that the results are beneficial. The above-discussed components namely support, confidence, and lift helps in measuring the strength of a rule to assist you in making an informed decision.

Market basket analysis example

Let's understand the market basket analysis with an example of where the Apriori algorithm is implemented in the arules package. It can be installed and run in R.

STEP 1: Catch a glance at the below-given table to understand how the data is loaded into the engine.

Transaction ID	Item name
1	Sandwich
1	Cookies
1	Milk
1	Egg
2	Burger
2	Pizza
2	Salad
2	Egg
2	Bottled water
2	Chicken
3	Sandwich
3	Egg
3	Bottled water

Source: Smartbridge

STEP 2: Next, each transaction is aggregated across records into a single record as an array that converts the data set to an R transaction. Check out the below image that depicts the result of the aggregation process.

Transaction ID	Items list
1	Cookies, Egg, Milk, Sandwich
2	Bottled water, Burger, Chicken, Egg, Pizza, Salad
3	Beacon, Bottled water, Egg, Sandwich, Yogurt
4	Burger, Pie, Pizza, Salad, Soda
5	Burger, ice cream, pie, pizza, salad, soda
6	Chocolate shake, cookies, egg, milk, sandwich
7	Beacon, chocolate shake, cookies, milk, yogurt
8	Bottled water, burger, chicken, chocolate shake, egg, pie, pizza, salad, soda
9	Beacon, bottled water, egg, milk, pizza, salad, yogurt
10	Chocolate shake, cookies, eggs, milk, sandwich
11	Beacon, burger, salad
12	Cookies, egg, milk, sandwich, yogurt
13	BEacon, bottled water, egg, pie, pizza, sandwich
14	Cookies, egg, milk, sandwich
15	Bottled water, burger, chicken, egg, pie, pizza, salad

STEP 3: Lastly, the Apriori logic is implemented in the transactions. Check the below-given image for the result set.

LHS	RHS	Rules	Support	Confidence	Lift
Ice cream	Soda	(Ice cream) -> (Soda)	0.07	1.00	5.00
Soda	Ice cream	(Soda) -> (Ice cream)	0.07	0.33	5.00
Ice cream	pie	(Ice cream) -> (Pie)	0.07	1.00	3.00
Pie	Ice cream	(Pie) -> (Ice cream)	0.07	0.20	3.00
Ice cream	Burger	(Ice cream) -> (Burger)	0.07	1.00	2.50
Burger	Ice cream	(Burger) -> (Ice cream)	0.07	0.17	2.50
Ice cream	Salad	(Ice cream) -> (Salad)	0.07	1.00	2.14
Salad	Ice cream	(Salad) -> (Ice cream)	0.07	0.14	2.14
Ice cream	Pizza	(Ice cream) -> (Pizza)	0.07	1.00	2.14
Pizza	Ice cream	(Pizza) -> (Ice cream)	0.07	0.14	2.14
Soda	Chicken	(Soda) -> (Chicken)	0.07	0.33	1.67
Chicken	Soda	(Chicken) -> (Soda)	0.07	0.33	1.67
Soda	Chocolate shake	(Soda) -> (Chocolate shake)	0.07	0.33	1.25
Chocolate shake	Soda	(Chocolate shake) -> (Soda)	0.07	0.25	1.25
Soda	Pie	(Soda) -> (Pie)	0.20	1.00	3.00
Pie	Soda	(Pie) -> (Soda)	0.20	0.60	3.00
Soda	Burger	(Soda) -> (Burger)	0.20	1.00	2.50
Burger	Soda	(Burger) -> (Soda)	0.20	0.50	2.50
Soda	Bottled water	(Soda) -> (Bottled water)	0.07	0.33	0.83
Bottled water	Soda	(Bottled water) -> (Soda	0.07	0.17	0.83
Soda	Salad	(Soda) -> (Salad)	0.20	1.00	2.14
Salad	Soda	(Salad) -> (Soda)	0.20	0.43	2.14
Soda	Pizza	(Soda) -> (Pizza)	0.20	1.00	2.14

The above image clearly depicts the number of strong consequent combinations in which soda is a keystone product category. Leveraging this information, the store manager can drive sales volume by keeping the price and margins low on soda.

Another eye-opening yet interesting result depicts that all the rules with ice cream illustrate the confidence of one with a significant lift. Thus, the store

manager can further promote ice cream with the belief that other items can be purchased by the customers with it at the same time.

Benefits of market basket analysis

The never-ending list of impeccable benefits that market basket analysis has to offer is widely being leveraged by organizations around the world. This is also the reason one can notice a spike in the hiring of ML engineers in companies around the world.

Read on to know some unmatched advantages that will leave you awestruck.

- For the retailers: Market basket analysis (MBA) finds great application for the marketing perspective to the retailers. It helps retailers in optimizing their marketing campaigns and strategize future sales by understanding their customers better. Offering actionable customer patterns and behavior helps in boosting the sales of the store and increases the ROI. Further, it enhances customer engagement and improves the customer experience while shopping from a store.
- Personalized recommendations: Market basket analysis understands the movie genre an individual likes to watch and recommends movies on OTT platforms like Amazon prime and Netflix.
- Promotions and campaigns: This analysis helps in understanding the products that go together along with the ones that form keystones in the product line.
- Customer behavior analysis: Market basket analysis helps you understand the customer behavior that is leveraged from UX/UI design to catalog design.
- In-store operations optimizations: MBA finds great application in optimizing the inventory by determining the popularity of the products in a store.
- New marketing tactics: MBA eases the task of discovering new marketing strategies to grow a brand. It analyses the demographic and gentrification data to help you make informed decisions on where to open your next store or target ads.

Applications of market basket analysis

The popularity of market basket analysis machine learning extends beyond the boundaries of the retail industry. We have listed some other areas where it is doing wonders below.

- Finance/Criminology: Market basket analysis finds great application in detecting fraud related to credit card usage data.
- Manufacturing: The predictive market basket analysis helps in predicting the failure of the equipment.
- Bioinformatics/Pharmaceutical: Market basket analysis helps in discovering the co-occurrence relationship among the pharmaceutically active ingredients and diagnosis that is prescribed to different groups of patients.
- Customer behavior: Using socio-economic and demographic data, market basket analysis helps in determining associated purchases.

- Medicine: Market basket analysis finds great use in determining symptom analysis and comorbid conditions in the medical field. It also helps in identifying the hereditary traits and genes that are associated with local environmental effects.
- Telecom: The telecom companies increasing attention to customer services is eased with market basket analysis. For instance, the companies in the telecom sector have started to deliver TV and internet packages together apart from the other discounted online services to eliminate churn.
- IBFS: IBFS organization leverages market basket analysis to trace the credit card history of a consumer. Based on the data received, such companies lower potential customers with attractive discounts on the go by employing sales professionals at shopping malls.

DATA COLLECTION

I collected data from 400 customers using a Questionnaire which will be shown below the report.

This report focuses on applying MBA to a dataset consisting of responses from 400 customers to a 300-question questionnaire.

The goal is to identify patterns and associations between different responses, which can be interpreted as relationships between various banking products and services.

Understanding the Dataset

Dataset Description

The dataset consists of 300 questions, each answered by 400 respondents. The questions cover various aspects of customer behavior and preferences, such as:

- Usage of banking products (e.g., savings accounts, loans, credit cards)
- Frequency of use of banking services (e.g., online banking, ATM visits)
- Customer satisfaction with different banking products
- Awareness and interest in additional banking services (e.g., insurance, mutual funds)

2. Question Types

The questions can be broadly categorized into the following types:

- **Yes/No Questions**: E.g., "Do you use online banking services?"
- **Satisfaction Scale**: E.g., "How satisfied are you with our credit card services? (1-5)"
- **Multiple Choice**: E.g., "Which banking products do you currently use?"
- **Open-Ended**: E.g., "Do you have any suggestions for improving our services?"

3. Data Preparation for Market Basket Analysis

3.1 Data Conversion

To perform MBA, the dataset needs to be transformed into a suitable format. For example:

- **Yes/No Questions**: Converted to binary format (1 for "Yes", 0 for "No").
- **Satisfaction Scale**: Responses of 4 or 5 are considered as "satisfied" and converted to binary.
- **Multiple Choice Questions**: Each option is treated as a separate "item" in the transaction.

3.2 Transaction Representation

Each respondent's answers represent a "transaction," with each "item" being a particular response to a question. This structure allows for the

application of MBA techniques, such as the Apriori algorithm, to identify frequent itemsets and association rules.

4. Market Basket Analysis Process

4.1 Application of the Apriori Algorithm

The Apriori algorithm was used to identify frequent itemsets within the dataset. The algorithm works by finding combinations of items that appear together in transactions more frequently than a specified minimum support threshold.

4.2 Generation of Association Rules

Once the frequent itemsets were identified, association rules were generated to explore the relationships between different responses. The key metrics used in this process were:

- **Support**: The proportion of transactions (respondents) containing a particular itemset.
- **Confidence**: The likelihood that a particular item (response) appears in a transaction given that another item is present.
- **Lift**: The strength of the association rule, indicating how much more likely the response is compared to random chance.

5. Results and Interpretation

5.1 Example Association Rules

Some example association rules derived from the analysis are:

- **Rule 1**: Customers who answered "Yes" to question 10 ("Are you aware of additional products?") are likely to answer "Yes" to question 11 ("Would you be interested in learning more?").
    - **Support**: 0.15 (15% of customers exhibit this behavior)
    - **Confidence**: 80% (80% of customers aware of additional products are interested in learning more)
    - **Lift**: 1.6 (The interest in learning more is 60% higher than random chance)
- **Rule 2**: Customers satisfied with their credit card services (scored 4 or 5 on satisfaction scale) are likely to also use a savings account.
    - **Support**: 0.25 (25% of customers exhibit this behavior)

- **Confidence**: 75% (75% of satisfied credit card users also use a savings account)
- **Lift**: 1.8 (This is 80% higher than random occurrence)

5.2 Insights

- **Cross-Selling Opportunities**: Customers who are aware of additional products are highly likely to be interested in learning more about them. This insight can guide targeted marketing efforts to inform and engage these customers about new offerings.
- **Product Bundling**: There is a strong association between satisfaction with credit card services and the usage of a savings account. This suggests that bundling these products or offering special deals for customers using both could enhance customer loyalty.

6. Conclusion

Market Basket Analysis has proven to be a valuable tool for identifying relationships between different banking products and customer behaviors. By understanding these associations, banks can:

- Develop targeted marketing strategies to promote relevant products to the right customers.
- Design product bundles that cater to customer preferences and usage patterns.
- Improve customer satisfaction by tailoring services and products to meet their needs.

The insights gained from this analysis can help the bank better understand its customers, enhance its product offerings, and ultimately drive business growth.

USING APRORI ALGORITHM

After the data has been collected in the form of questionnaire the data is then entered into the spreadsheet and aprori Algorithm is applied to it.

+ Code + Text

38s

```python
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

# Load the dataset
df = pd.read_excel('Hypothetical_Responses_300_Questions_400_Customers.xlsx')

# Convert Yes/No answers to binary format
df_binary = df.applymap(lambda x: 1 if x == 'Yes' else 0)

# Apply the Apriori algorithm to find frequent itemsets
frequent_itemsets = apriori(df_binary, min_support=0.1, use_colnames=True)

# Generate association rules
rules = association_rules(frequent_itemsets, metric="lift", min_threshold=1.0)

# Sort rules by confidence
rules = rules.sort_values(by='confidence', ascending=False)

# Save the results to a file
rules.to_excel('Market_Basket_Analysis_Results.xlsx', index=False)

# Display the top 5 rules
print(rules.head())
```

```
<ipython-input-1-ffc066754e72>:8: FutureWarning: DataFrame.applymap has been deprecat
  df_binary = df.applymap(lambda x: 1 if x == 'Yes' else 0)
/usr/local/lib/python3.10/dist-packages/mlxtend/frequent_patterns/fpcommon.py:109: De
  warnings.warn(
                antecedents consequents  antecedent support  \
117692  (Q76, Q256, Q231)       (Q151)              0.1425
```

Connected to Python 3 Google Compute Engine backend

OUTPUT OF THE CODE

```
<ipython-input-1-ffc066754e72>:8: FutureWarning: DataFrame.applymap has been deprecatec
  df_binary = df.applymap(lambda x: 1 if x == 'Yes' else 0)
/usr/local/lib/python3.10/dist-packages/mlxtend/frequent_patterns/fpcommon.py:109: Depr
  warnings.warn(
              antecedents consequents  antecedent support  \
117692   (Q76, Q256, Q231)      (Q151)              0.1425
119694  (Q116, Q231, Q241)      (Q201)              0.1225
107617   (Q81, Q201, Q166)       (Q11)              0.1300
109228  (Q266, Q131, Q166)       (Q11)              0.1275
112477    (Q41, Q236, Q61)      (Q116)              0.1250

        consequent support  support  confidence      lift  leverage  \
117692               0.515   0.1175    0.824561  1.601090  0.044112
119694               0.550   0.1000    0.816327  1.484230  0.032625
107617               0.540   0.1050    0.807692  1.495726  0.034800
109228               0.540   0.1025    0.803922  1.488744  0.033650
112477               0.535   0.1000    0.800000  1.495327  0.033125

        conviction  zhangs_metric
117692      2.7645       0.437814
119694      2.4500       0.371795
107617      2.3920       0.380952
109228      2.3460       0.376267
112477      2.3250       0.378571
```

INTREPRETATION FROM OUTPUT OF THE CODE AND SUMMARY OF THE FINDING

Interpretation

Association Rules

Here are some association rules derived from the analysis:

- **Rule 1**: Customers who answered "Yes" to question 76 ("Do you use mobile banking services?") and question 231 ("Do you have a savings account with us?") are likely to answer "Yes" to question 151 ("Are you satisfied with the mobile banking app?").
    - **Support**: 0.12 (12% of customers exhibit this behavior)
    - **Confidence**: 82.46% (82.46% of customers who use mobile banking and have a savings account are satisfied with the mobile banking app)
    - **Lift**: 1.60 (The satisfaction with the mobile banking app is 60% higher than random chance)

  Interpretation: This rule indicates that customers who engage with mobile banking services and have a savings account are significantly more likely to be satisfied with the bank's mobile banking app. This suggests that customers who are already using the bank's mobile services have a favorable view of its functionality and performance.

- **Rule 2**: Customers who answered "Yes" to question 116 ("Do you use our credit card services?") and question 241 ("Do you currently have a loan with us?") are likely to answer "Yes" to question 201 ("Do you plan to open a fixed deposit account within the next year?").
    - **Support**: 0.10 (10% of customers exhibit this behavior)
    - **Confidence**: 81.63% (81.63% of customers who use credit cards and have a loan also plan to open a fixed deposit account)
    - **Lift**: 1.48 (The intention to open a fixed deposit account is 48% higher than random chance)

Interpretation: This rule suggests a strong relationship between the usage of credit card services, holding a loan, and planning to open a fixed deposit account. Customers who are already utilizing credit cards and loans are more inclined to invest further by opening fixed deposit accounts, presenting a cross-selling opportunity for the bank.

- **Rule 3**: Customers who answered "Yes" to question 81 ("Have you taken a mortgage loan from our bank?"), question 201 ("Do you plan to open a fixed deposit account within the next year?"), and question 166 ("Do you contribute to our retirement savings plan?") are likely to answer "Yes" to question 11 ("Would you recommend our bank to your family and friends?").
    - **Support**: 0.13 (13% of customers exhibit this behavior)
    - **Confidence**: 80.76% (80.76% of customers who have a mortgage loan, plan to open a fixed deposit account, and contribute to a retirement savings plan are likely to recommend the bank)
    - **Lift**: 1.50 (The likelihood of recommending the bank is 50% higher than random chance)

  Interpretation: Customers who are deeply involved with multiple significant financial products—such as mortgage loans, fixed deposits, and retirement savings—are more likely to recommend the bank. This indicates a strong positive correlation between high engagement with key products and customer loyalty.

Suggestions:

Enhance Mobile Banking Experience

Background: The analysis revealed that customers who use mobile banking services and have savings accounts are more likely to be satisfied with the bank's mobile app. This positive correlation highlights the importance of the mobile banking experience in overall customer satisfaction.

Strategy:

User Experience (UX) Improvement: Invest in continuous improvement of the mobile banking app's user interface and experience. Regularly conduct user experience testing and gather feedback from customers to identify pain points and areas for enhancement.

Feature Expansion: Introduce new features based on customer feedback, such as personalized notifications, easier navigation, and additional security measures. Providing features like expense tracking, bill reminders, and financial planning tools can further increase engagement.

Educational Campaigns: Launch educational campaigns to ensure that customers are aware of the app's full capabilities. Tutorials, webinars, and in-app guides can help users fully utilize the mobile banking app, leading to higher satisfaction.

Regular Updates and Maintenance: Ensure the mobile app is regularly updated to fix bugs, improve speed, and introduce new functionalities. Highlight these updates to users to show the bank's commitment to enhancing their digital experience.

Expected Impact: By enhancing the mobile banking experience, the bank can increase customer satisfaction, encourage more customers to adopt mobile banking, and potentially reduce the operational load on physical branches.

Cross-Sell Fixed Deposit Accounts

Background: Customers who use credit card services and have existing loans are more likely to be interested in fixed deposit accounts. This insight presents a significant opportunity for cross-selling.

Strategy:

Personalized Offers: Develop personalized marketing campaigns targeting customers who hold both credit cards and loans. Use data-driven insights to create tailored messages that highlight the benefits of fixed deposits, such as guaranteed returns and safety of investment.

Bundling Products: Create bundled financial products that combine credit cards, loans, and fixed deposits. Offering incentives like better interest rates on fixed deposits for customers who hold multiple products can increase uptake.

Automated Suggestions: Implement automated suggestions in online banking portals and mobile apps that recommend fixed deposit accounts to customers who have credit cards and loans. For example, when a customer logs in to pay their credit card bill, the system could suggest a fixed deposit based on their financial behavior.

Relationship Manager Outreach: Train relationship managers to identify and approach customers who are likely candidates for fixed deposits based on their current product usage. A personal touch from a knowledgeable advisor can increase conversion rates.

Expected Impact: Effective cross-selling can lead to higher product penetration rates, increased customer loyalty, and a more diversified revenue stream for the bank.

Leverage High Engagement for Referrals

Background: The analysis shows that customers deeply engaged with multiple banking products are more likely to recommend the bank to others. This finding underscores the potential of referral programs in driving customer acquisition.

Strategy:

Referral Incentive Programs: Design a referral program that rewards both the referrer and the new customer. Offer attractive incentives such as cash rewards, bonus interest rates, or exclusive product offers for successful referrals.

Targeted Communication: Use data analytics to identify highly engaged customers who hold multiple products, and target them with personalized referral campaign emails or in-app messages. Emphasize the ease of referring friends and the benefits they both stand to gain.

Referral Tracking Tools: Develop easy-to-use referral tracking tools within the bank's digital platforms. Customers should be able to refer friends through the app or website and track the status of their referrals.

Social Media Integration: Integrate referral programs with social media platforms to encourage customers to share their positive experiences with the bank. Provide easy sharing options and incentives for social media referrals.

Expected Impact: By leveraging the existing goodwill of highly engaged customers, the bank can expand its customer base at a relatively low cost while simultaneously increasing the lifetime value of current customers.

Focus on Comprehensive Financial Solutions

Background: Customers who are engaged with multiple financial products are more loyal and satisfied. Offering comprehensive financial solutions can capitalize on this trend.

Strategy:

Product Bundling: Develop and promote product bundles that address different life stages and financial goals. For example, a "Family Financial Plan" could include a mortgage, savings accounts for children, and a retirement plan. These bundles should be marketed as convenient, all-in-one solutions for managing finances.

Customer Journey Mapping: Create detailed customer journey maps to understand the typical path that leads customers to purchase multiple products. Use these insights to design marketing and sales strategies that guide customers through a progressive engagement with the bank's offerings.

Loyalty Programs: Introduce or enhance loyalty programs that reward customers for using multiple products. Points, discounts on fees, or special rates for loyal customers can incentivize them to stay within the bank's ecosystem.

Financial Wellness Programs: Offer financial wellness programs that educate customers on managing their finances effectively. These programs could be tied to product recommendations, helping customers see the value in comprehensive financial solutions.

Expected Impact: Focusing on comprehensive financial solutions will likely lead to higher customer retention, greater cross-selling opportunities, and a stronger relationship between the bank and its customers.

Targeted Marketing Campaigns

Background: The market basket analysis provides insights into specific customer segments that are more likely to purchase additional products based on their current usage.

Strategy:

Segment-Specific Campaigns: Develop marketing campaigns tailored to the needs and preferences of specific customer segments identified in the analysis. For instance, customers who frequently use online banking might be more receptive to digital-only offers or products.

Behavioral Triggers: Use behavioral triggers to initiate marketing campaigns. For example, if a customer frequently uses the bank's credit card but does not have a loan, the system could automatically trigger a loan offer.

Omnichannel Approach: Ensure that marketing campaigns are consistent across all channels—online, mobile, email, and in-branch. This consistency reinforces the message and makes it easier for customers to respond, regardless of how they interact with the bank.

Data-Driven Insights: Continuously analyze customer data to refine marketing strategies. Use the insights from market basket analysis to identify emerging trends and adjust campaigns accordingly.

Expected Impact: Targeted marketing campaigns, informed by data insights, are likely to achieve higher conversion rates, improve customer engagement, and lead to more efficient use of marketing resources.

Summary of Recommendations

In summary, the analysis suggests that the bank can significantly enhance customer satisfaction and loyalty by focusing on improving the mobile banking experience, effectively cross-selling related products, leveraging customer referrals, offering comprehensive financial solutions, and executing targeted marketing campaigns. By implementing these strategies, the bank can strengthen its customer relationships, increase product penetration, and ultimately, drive long-term growth

Conclusion

The internship at J&K Bank provided a comprehensive opportunity to explore the application of Market Basket Analysis (MBA) within the banking sector. The primary objective was to identify relationships between different banking products based on customer behavior and preferences, thereby providing actionable insights for strategic decision-making.

Through the analysis of hypothetical data gathered from 400 customers, several key associations were identified. These insights revealed that customers who use certain banking products are more likely to engage with others, offering a significant opportunity for cross-selling and upselling. For example, customers satisfied with their credit card services are more likely to use savings accounts, and those aware of additional products are inclined to learn more, underscoring the potential for targeted marketing and product bundling.

The project also highlighted the importance of enhancing the mobile banking experience, as a significant portion of customer satisfaction is linked to the usability and features of the bank's digital platforms. Additionally, the referral behavior of highly engaged customers presented an avenue for organic growth through customer advocacy.

The recommendations provided in this report, including improving mobile banking, cross-selling fixed deposits, leveraging high engagement for referrals, focusing on comprehensive financial solutions, and executing targeted marketing campaigns, are designed to capitalize on the insights gained from the MBA. These strategies, when implemented effectively, have the potential to

significantly enhance customer satisfaction, increase product penetration, and strengthen the bank's competitive position in the market.

In conclusion, the Market Basket Analysis conducted during this internship has demonstrated the value of data-driven decision-making in the banking industry. The insights derived from customer behavior can be leveraged to develop more personalized and effective marketing strategies, ultimately leading to better customer engagement and business growth. This internship experience has provided a solid foundation in applying analytical techniques to real-world business challenges, equipping me with valuable skills and knowledge that will be beneficial in my future career endeavors.

QUESTIONNAIRE

Questionnaire file consisted of 40 pages that’s why snap shots from the file are being uploaded.

Customer Questionnaire for Market Basket Analysis in Banking

Dear Customer,

We are conducting a study to better understand how our customers use different banking products and services. Your responses will help us improve our offerings and tailor them to better suit your needs. Please take a few minutes to answer the following questions. Your responses will be kept confidential.

Thank you for your time!

1. 1. How long have you been a customer of our bank?

- A) Less than 1 year
- B) 1-3 years
- C) 3-5 years
- D) More than 5 years

2. 2. Which of the following banking products do you currently use? (You can select more than one)

- A) Savings Account
- B) Current Account
- C) Fixed Deposit
- D) Credit Card
- E) Personal Loan
- F) Home Loan
- G) Insurance

3. 3. How frequently do you use online banking services?

- A) Daily
- B) Weekly
- C) Monthly
- D) Rarely

4. 4. Which product do you use most frequently?

- A) Savings Account
- B) Credit Card
- C) Online Transfers
- D) Fixed Deposits

5. 5. Have you ever availed a loan from our bank?

- A) Yes
- B) No

6. 6. If yes, what type of loan did you avail?

- A) Personal Loan
- B) Home Loan
- C) Car Loan
- D) Education Loan

7. 7. How often do you use the following services? (Rate on a scale of 1 to 5, where 1 is Never and 5 is Always)

- A) Online Banking
- B) ATM Services
- C) Branch Visits
- D) Mobile Banking

8. 8. Do you hold a credit card with our bank?

- A) Yes
- B) No

9. 9. How satisfied are you with our credit card services?

- A) Very Satisfied
- B) Satisfied
- C) Neutral
- D) Dissatisfied
- E) Very Dissatisfied

10. 10. Are you aware of the additional products our bank offers (e.g. insurance, mutual funds)?

- A) Yes
- B) No

11. 11. Would you be interested in learning more about these products?

- A) Yes
- B) No

12. 12. Have you purchased any insurance products through our bank?

- A) Yes
- B) No

13. 13. If yes, which type of insurance did you purchase?

- A) Life Insurance
- B) Health Insurance
- C) Vehicle Insurance
- D) Home Insurance

14. 14. Are you considering any of the following services in the near future? (You can select more than one)

- A) Applying for a loan
- B) Opening a fixed deposit
- C) Investing in mutual funds
- D) Purchasing insurance

15. 15. How do you usually learn about our new products and services?

- A) Bank's Website

31. 15. How do you usually learn about our new products and services?

- A) Bank's Website
- B) Email Notifications
- C) SMS Alerts
- D) Branch Visits
- E) Social Media

32. 16. On a scale of 1 to 5, how likely are you to recommend our bank to a friend or colleague?

- A) 1 - Not Likely
- B) 2
- C) 3
- D) 4
- E) 5 - Very Likely

33. 17. Do you have any suggestions for improving our products and services?

- [Open-ended response]

34. 1. How long have you been a customer of our bank?

- A) Less than 1 year
- B) 1-3 years
- C) 3-5 years
- D) More than 5 years

35. 2. Which of the following banking products do you currently use? (You can select more than one)

- A) Savings Account
- B) Current Account
- C) Fixed Deposit
- D) Credit Card
- E) Personal Loan
- F) Home Loan
- G) Insurance

37. 3. How frequently do you use online banking services?

- A) Daily
- B) Weekly
- C) Monthly
- D) Rarely

38. 4. Which product do you use most frequently?

- A) Savings Account
- B) Credit Card
- C) Online Transfers
- D) Fixed Deposits

39. 5. Have you ever availed a loan from our bank?

- A) Yes
- B) No

40. 6. If yes, what type of loan did you avail?

- A) Personal Loan
- B) Home Loan
- C) Car Loan
- D) Education Loan

41. 7. How often do you use the following services? (Rate on a scale of 1 to 5, where 1 is Never and 5 is Always)

- A) Online Banking
- B) ATM Services
- C) Branch Visits
- D) Mobile Banking

42. 8. Do you hold a credit card with our bank?

- A) Yes
- B) No

43. 9. How satisfied are you with our credit card services?

- A) Very Satisfied
- B) Satisfied
- C) Neutral
- D) Dissatisfied
- E) Very Dissatisfied

44. 10. Are you aware of the additional products our bank offers (e.g. insurance, mutual funds)?

- A) Yes
- B) No

45. 11. Would you be interested in learning more about these products?

- A) Yes
- B) No

46. 12. Have you purchased any insurance products through our bank?

- A) Yes
- B) No

47. 13. If yes, which type of insurance did you purchase?

- A) Life Insurance
- B) Health Insurance
- C) Vehicle Insurance
- D) Home Insurance

64. 13. If yes, which type of insurance did you purchase?

- A) Life Insurance
- B) Health Insurance
- C) Vehicle Insurance
- D) Home Insurance

65. 14. Are you considering any of the following services in the near future? (You can select more than one)

- A) Applying for a loan
- B) Opening a fixed deposit
- C) Investing in mutual funds
- D) Purchasing insurance

66. 15. How do you usually learn about our new products and services?

- A) Bank's Website
- B) Email Notifications
- C) SMS Alerts
- D) Branch Visits
- E) Social Media

67. 16. On a scale of 1 to 5, how likely are you to recommend our bank to a friend or colleague?

- A) 1 - Not Likely
- B) 2
- C) 3
- D) 4
- E) 5 - Very Likely

68. 17. Do you have any suggestions for improving our products and services?

- (Open-ended response)

69. 1. How long have you been a customer of our bank?

- A) Less than 1 year
- B) 1-3 years
- C) 3-5 years
- D) More than 5 years

70. 2. Which of the following banking products do you currently use? (You can select more than one)

- A) Savings Account
- B) Current Account
- C) Fixed Deposit
- D) Credit Card
- E) Personal Loan
- F) Home Loan
- G) Insurance

71. 3. How frequently do you use online banking services?

- A) Daily
- B) Weekly
- C) Monthly
- D) Rarely

72. 4. Which product do you use most frequently?

- A) Savings Account
- B) Credit Card
- C) Online Transfers
- D) Fixed Deposits

73. 5. Have you ever availed a loan from our bank?

- A) Yes
- B) No

74. 6. If yes, what type of loan did you avail?

- A) Personal Loan
- B) Home Loan
- C) Car Loan
- D) Education Loan

75. 7. How often do you use the following services? (Rate on a scale of 1 to 5, where 1 is Never and 5 is Always)

- A) Online Banking
- B) ATM Services
- C) Branch Visits
- D) Mobile Banking

76. 8. Do you hold a credit card with our bank?

- A) Yes
- B) No

77. 9. How satisfied are you with our credit card services?

- A) Very Satisfied
- B) Satisfied
- C) Neutral
- D) Dissatisfied
- E) Very Dissatisfied

78. 10. Are you aware of the additional products our bank offers (e.g., insurance, mutual funds)?

- A) Yes
- B) No

79. 11. Would you be interested in learning more about these products?

- A) Yes
- B) No

177. 7. How often do you use the following services? (Rate on a scale of 1 to 5, where 1 is Never and 5 is Always)

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- B) No

181. 11. Would you be interested in learning more about these products?

- A) Yes
- B) No

182. 12. Have you purchased any insurance products through our bank?

- A) Yes
- B) No

183. 13. If yes, which type of insurance did you purchase?

- A) Life Insurance
- B) Health Insurance
- C) Vehicle Insurance
- D) Home Insurance

184. 14. Are you considering any of the following services in the near future? (You can select more than one)

- A) Applying for a loan
- B) Opening a fixed deposit
- C) Investing in mutual funds
- D) Purchasing insurance

185. 15. How do you usually learn about our new products and services?

- A) Bank's Website
- B) Email Notifications
- C) SMS Alerts
- D) Branch Visits
- E) Social Media

186. 16. On a scale of 1 to 5, how likely are you to recommend our bank to a friend or colleague?

- A) 1 - Not Likely
- B) 2
- C) 3
- D) 4
- E) 5 - Very Likely

187. 17. Do you have any suggestions for improving our products and services?

- (Open-ended response)

188. 1. How long have you been a customer of our bank?

- A) Less than 1 year
- B) 1-3 years
- C) 3-5 years
- D) More than 5 years

189. 2. Which of the following banking products do you currently use? (You can select more than one)

- A) Savings Account
- B) Current Account
- C) Fixed Deposit
- D) Credit Card
- E) Personal Loan
- F) Home Loan
- G) Insurance

190. 3. How frequently do you use online banking services?

- A) Daily
- B) Weekly
- C) Monthly
- D) Rarely

191. 4. Which product do you use most frequently?

- A) Savings Account
- B) Credit Card
- C) Online Transfers
- D) Fixed Deposits

192. 5. Have you ever availed a loan from our bank?

- A) Yes
- B) No

274. 2. Which of the following banking products do you currently use? (You can select more than one)

- A) Savings Account
- B) Current Account
- C) Fixed Deposit
- D) Credit Card
- E) Personal Loan
- F) Home Loan
- G) Insurance

275. 3. How frequently do you use online banking services?

- A) Daily
- B) Weekly
- C) Monthly
- D) Rarely

276. 4. Which product do you use most frequently?

- A) Savings Account
- B) Credit Card
- C) Online Transfers
- D) Fixed Deposits

277. 5. Have you ever availed a loan from our bank?

- A) Yes
- B) No

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- (Open-ended response)